AI-Powered Satellite Attitude Determination and Control System (Hybrid ADCS)

Project Report

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Abstract

Attitude Determination and Control Systems (ADCS) are essential subsystems of modern satellites, responsible for maintaining the desired spacecraft orientation during mission operations. Small satellites and CubeSats are particularly vulnerable to environmental disturbances, sensor uncertainties, and actuator limitations, which can degrade pointing accuracy and mission performance.

This project presents an AI-Powered Satellite Attitude Determination and Control System (Hybrid ADCS) that combines classical estimation and control methods with machine learning-based control augmentation. A quaternion-based spacecraft dynamics model is developed to simulate three-axis rotational motion under external disturbance torques. Attitude estimation is performed using a Multiplicative Extended Kalman Filter (MEKF), which fuses noisy sensor measurements and estimates spacecraft orientation and angular velocity.

A hybrid control architecture is implemented consisting of conventional PD and LQR controllers together with an AI-based neural controller that provides adaptive disturbance compensation. The proposed system is validated through multiple simulation scenarios including attitude stabilization, AI-versus-LQR performance comparison, Monte Carlo robustness analysis, sensor failure recovery experiments, and multi-orbit mission simulations.

Results demonstrate that the hybrid ADCS framework successfully maintains spacecraft stability under disturbances while providing a flexible platform for evaluating AI-assisted spacecraft control strategies. The project highlights the potential of integrating artificial intelligence with traditional spacecraft Guidance, Navigation and Control (GNC) systems for future autonomous satellite missions.

CHAPTER 1

INTRODUCTION

1.1 Background

Satellites play a critical role in modern communication, Earth observation, navigation, scientific exploration, defense, and space research missions. For a satellite to successfully accomplish its mission objectives, it must maintain a desired orientation in space. The subsystem responsible for determining and controlling spacecraft orientation is known as the Attitude Determination and Control System (ADCS).

The ADCS continuously estimates the satellite attitude using onboard sensors and generates control commands to actuators such as reaction wheels, magnetorquers, or control moment gyroscopes. Accurate attitude control is essential for payload pointing, antenna alignment, solar panel orientation, and mission stability.

Traditional spacecraft attitude control systems rely on estimation algorithms such as Kalman Filters and control methods such as Proportional-Derivative (PD) and Linear Quadratic Regulator (LQR) controllers. These approaches provide reliable performance under nominal operating conditions but often experience performance degradation when subjected to nonlinear disturbances, sensor uncertainties, and model inaccuracies.

Recent advances in Artificial Intelligence (AI) and Machine Learning (ML) have created opportunities to enhance spacecraft autonomy. AI-based controllers can learn complex nonlinear relationships and compensate for unmodeled disturbances that may be difficult to capture using conventional mathematical models. Integrating AI with classical spacecraft control architectures has emerged as an active research area in modern Guidance, Navigation, and Control (GNC) systems.

This project investigates the development of a Hybrid Artificial Intelligence Powered Attitude Determination and Control System that combines classical estimation and control techniques with neural network-based disturbance compensation for improved spacecraft attitude stabilization.

1.2 Problem Statement

Small satellites and CubeSats operate in harsh space environments where various disturbance torques continuously affect spacecraft orientation. These disturbances include gravity gradient effects, solar radiation pressure, atmospheric drag, actuator imperfections, and sensor noise.

Traditional control methods such as PD and LQR controllers are effective when system dynamics are accurately modeled. However, real spacecraft often encounter uncertainties and nonlinear effects that cannot be completely represented using simplified mathematical models. These limitations can result in degraded pointing accuracy, increased control effort, and slower disturbance recovery.

Additionally, sensor failures and temporary measurement outages may significantly affect attitude estimation performance. Ensuring robust operation during such events is an important requirement for modern autonomous spacecraft.

Therefore, there is a need for an intelligent control framework capable of:

- Maintaining stable spacecraft attitude under disturbances.
- Improving pointing accuracy.
- Enhancing robustness against sensor uncertainties.
- Recovering from temporary sensor failures.

- Supporting long-duration autonomous operation.

This project addresses these challenges through a hybrid architecture that combines a Multiplicative Extended Kalman Filter (MEKF) with AI-assisted spacecraft attitude control.

1.3 Objectives

The primary objective of this project is to develop and validate a Hybrid Artificial Intelligence Powered Attitude Determination and Control System for small satellites.

The specific objectives are:

1. Develop a three-axis spacecraft attitude dynamics simulation using quaternion representation.
2. Implement realistic spacecraft sensor models including gyroscope and reference vector measurements.
3. Design and implement a Multiplicative Extended Kalman Filter (MEKF) for attitude estimation and sensor fusion.
4. Develop classical spacecraft attitude controllers including PD and LQR control methods.
5. Implement an AI-based neural controller for disturbance compensation and control augmentation.
6. Compare the performance of classical and AI-assisted control approaches.
7. Evaluate robustness under disturbance conditions using Monte Carlo simulations.
8. Investigate sensor failure recovery capability during temporary measurement outages.

9. Perform long-duration multi-orbit simulations to evaluate mission-level performance.

10. Generate a complete validation framework suitable for aerospace research and advanced academic applications.

1.4 Project Contributions

The major contributions of this work are summarized as follows:

Contribution 1: Quaternion-Based Spacecraft Dynamics Simulator

A complete three-axis spacecraft rotational dynamics simulator was developed using quaternion kinematics and rigid-body dynamics equations.

Contribution 2: Multiplicative Extended Kalman Filter

A Multiplicative Extended Kalman Filter was implemented to estimate spacecraft attitude and angular velocity while accounting for sensor noise and bias.

Contribution 3: Hybrid AI-Based Attitude Controller

An AI-assisted control framework was developed by augmenting conventional controllers with a neural network-based control component capable of disturbance compensation.

Contribution 4: AI versus Classical Controller Comparison

A dedicated validation framework was created to compare AI-assisted control performance against traditional LQR-based control approaches.

Contribution 5: Sensor Failure Recovery Analysis

The system was evaluated under simulated Star Tracker outages to assess estimation and control robustness during sensor failures.

Contribution 6: Multi-Orbit Mission Simulation

A long-duration mission simulation framework was implemented to evaluate spacecraft performance over multiple orbital periods.

Contribution 7: Comprehensive Validation Framework

Monte Carlo analysis, diagnostic testing, estimation validation, disturbance testing, and mission-level simulations were integrated into a unified validation framework.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Attitude Determination and Control Systems (ADCS) are among the most critical subsystems of modern spacecraft. Their primary function is to estimate and control spacecraft orientation to ensure proper payload operation, communication antenna alignment, power generation, and mission success. Over the past several decades, significant research has been conducted in spacecraft attitude estimation, control theory, sensor fusion, and autonomous spacecraft guidance.

Traditional ADCS architectures rely on mathematical models of spacecraft dynamics together with estimation and control algorithms. Recent developments in Artificial Intelligence (AI) have introduced new possibilities for improving spacecraft autonomy and robustness under uncertain operating conditions.

This chapter reviews the fundamental concepts and previous research related to spacecraft attitude determination, Kalman filtering, classical attitude control techniques, and AI-based spacecraft control systems.

2.2 Spacecraft Attitude Determination and Control Systems

A spacecraft operates in a three-dimensional environment and must continuously maintain a desired orientation relative to inertial space or mission targets. The process of estimating spacecraft orientation is known as attitude determination, while generating control actions to achieve the desired orientation is known as attitude control.

Typical spacecraft ADCS architectures consist of:

- Sensors
- Estimators
- Controllers
- Actuators

Sensors measure spacecraft motion and orientation. Estimators process sensor data to determine spacecraft attitude. Controllers generate control commands, and actuators apply the required torques to achieve desired spacecraft motion.

The ADCS must operate continuously throughout a mission while maintaining stability, robustness, and accuracy under external disturbances and hardware limitations.

2.3 Spacecraft Attitude Representation

Several mathematical approaches are used to represent spacecraft attitude.

Euler Angles

Euler angles describe orientation through three sequential rotations. Although intuitive, Euler angles suffer from singularities known as gimbal lock.

Direction Cosine Matrices (DCM)

Direction Cosine Matrices represent attitude using a 3×3 orthogonal rotation matrix. DCMs avoid singularities but require nine parameters, increasing computational complexity.

Quaternions

Quaternions provide a compact and singularity-free representation of spacecraft attitude using four parameters. Due to their numerical stability and computational efficiency, quaternions have become the preferred attitude representation in modern spacecraft ADCS implementations.

Consequently, quaternion-based attitude representation is adopted in this project.

2.4 Kalman Filtering for Spacecraft Attitude Estimation

Accurate attitude estimation requires combining measurements from multiple sensors while accounting for measurement noise and uncertainties.

The Kalman Filter, introduced by Rudolf Kalman in 1960, provides an optimal framework for state estimation in linear systems. However, spacecraft attitude dynamics are inherently nonlinear.

To address nonlinearities, several variants of the Kalman Filter have been developed:

Extended Kalman Filter (EKF)

The Extended Kalman Filter linearizes nonlinear system equations around the current operating point. EKFs have been widely used in aerospace applications for navigation and attitude estimation.

Unscented Kalman Filter (UKF)

The Unscented Kalman Filter approximates nonlinear state propagation using sigma points rather than linearization. UKFs often provide improved estimation accuracy at the expense of increased computational complexity.

Multiplicative Extended Kalman Filter (MEKF)

The Multiplicative Extended Kalman Filter is specifically designed for quaternion-based attitude estimation. Instead of directly estimating quaternions, the MEKF estimates attitude error states, preserving quaternion normalization and improving numerical stability.

Because of its suitability for spacecraft attitude determination, the MEKF is implemented in this project.

2.5 Classical Spacecraft Attitude Control

Spacecraft attitude control aims to generate actuator commands that minimize attitude error while maintaining system stability.

Proportional-Derivative (PD) Control

PD control is one of the most widely used spacecraft attitude control techniques due to its simplicity and reliability.

The proportional component reduces attitude error, while the derivative component provides damping and suppresses oscillations.

Advantages include:

- Simple implementation
- Low computational requirements
- Good stability characteristics

Limitations include:

- Sensitivity to disturbances
- Limited adaptability to changing operating conditions

Linear Quadratic Regulator (LQR)

The Linear Quadratic Regulator is an optimal control technique that minimizes a cost function balancing attitude error and control effort.

Advantages include:

- Optimal performance
- Improved disturbance rejection
- Efficient control effort management

However, LQR performance depends heavily on the accuracy of the mathematical model used during controller design.

2.6 Artificial Intelligence in Spacecraft Control

Recent advances in Artificial Intelligence and Machine Learning have created opportunities for improving spacecraft autonomy.

AI-based controllers can learn complex nonlinear relationships from data and compensate for disturbances that may be difficult to model mathematically.

Applications of AI in spacecraft systems include:

- Attitude control
- Autonomous navigation

- Fault detection
- Mission planning
- Spacecraft health monitoring
- Guidance and trajectory optimization

Neural networks are among the most commonly used AI models because they can approximate highly nonlinear functions and adapt to varying system conditions.

Several studies have demonstrated the ability of neural networks to improve spacecraft control performance under uncertain environmental conditions.

2.7 Hybrid AI-Based Spacecraft Control

Rather than replacing classical controllers entirely, recent research has focused on hybrid control architectures that combine traditional control theory with AI techniques.

In hybrid control systems:

- Classical controllers provide stability and reliability.
- AI components provide adaptive disturbance compensation.
- Estimation algorithms supply accurate state information.

This approach combines the strengths of both methodologies while reducing their individual limitations.

Hybrid architectures are particularly attractive for CubeSat missions because they offer improved performance without requiring highly complex onboard computational resources.

The present project follows this philosophy by integrating a neural network-based controller with classical spacecraft attitude control algorithms.

2.8 Sensor Failures and Fault Tolerance

Spacecraft sensors may experience temporary failures caused by:

- Radiation effects
- Sun blinding
- Hardware degradation
- Communication interruptions

A robust ADCS must continue operating safely during such events.

Fault-tolerant spacecraft systems typically rely on:

- Redundant sensors
- State estimation algorithms
- Recovery strategies

The ability of an ADCS to recover from temporary sensor outages is an important indicator of mission robustness.

Therefore, sensor failure recovery analysis is included as part of this project's validation framework.

2.9 Long-Duration Spacecraft Operations

Many spacecraft missions require continuous operation for months or years.

Consequently, ADCS performance must be evaluated not only during short simulations but also during long-duration mission scenarios.

Multi-orbit simulations provide valuable insight into:

- Long-term stability
- Disturbance accumulation
- Actuator usage
- Estimator consistency
- Mission robustness

For this reason, a multi-orbit mission simulation framework was developed and evaluated in this project.

2.10 Summary

This chapter reviewed the fundamental concepts underlying spacecraft attitude determination and control systems. Previous research on attitude representation, Kalman filtering, classical control methods, artificial intelligence, fault tolerance, and long-duration spacecraft operation was discussed.

The review highlights the growing importance of hybrid AI-assisted control architectures that combine traditional estimation and control techniques with machine learning-based disturbance compensation.

Building upon these concepts, the next chapter presents the overall architecture of the proposed Hybrid AI-Powered Satellite Attitude Determination and Control System.

CHAPTER 3

SYSTEM ARCHITECTURE

3.1 Introduction

The proposed Hybrid Artificial Intelligence Powered Attitude Determination and Control System (ADCS) integrates spacecraft dynamics, sensor models, state estimation algorithms, classical control methods, and AI-based control augmentation into a unified simulation framework.

The objective of the architecture is to maintain accurate spacecraft attitude despite the presence of disturbances, sensor noise, estimation uncertainty, and temporary sensor failures.

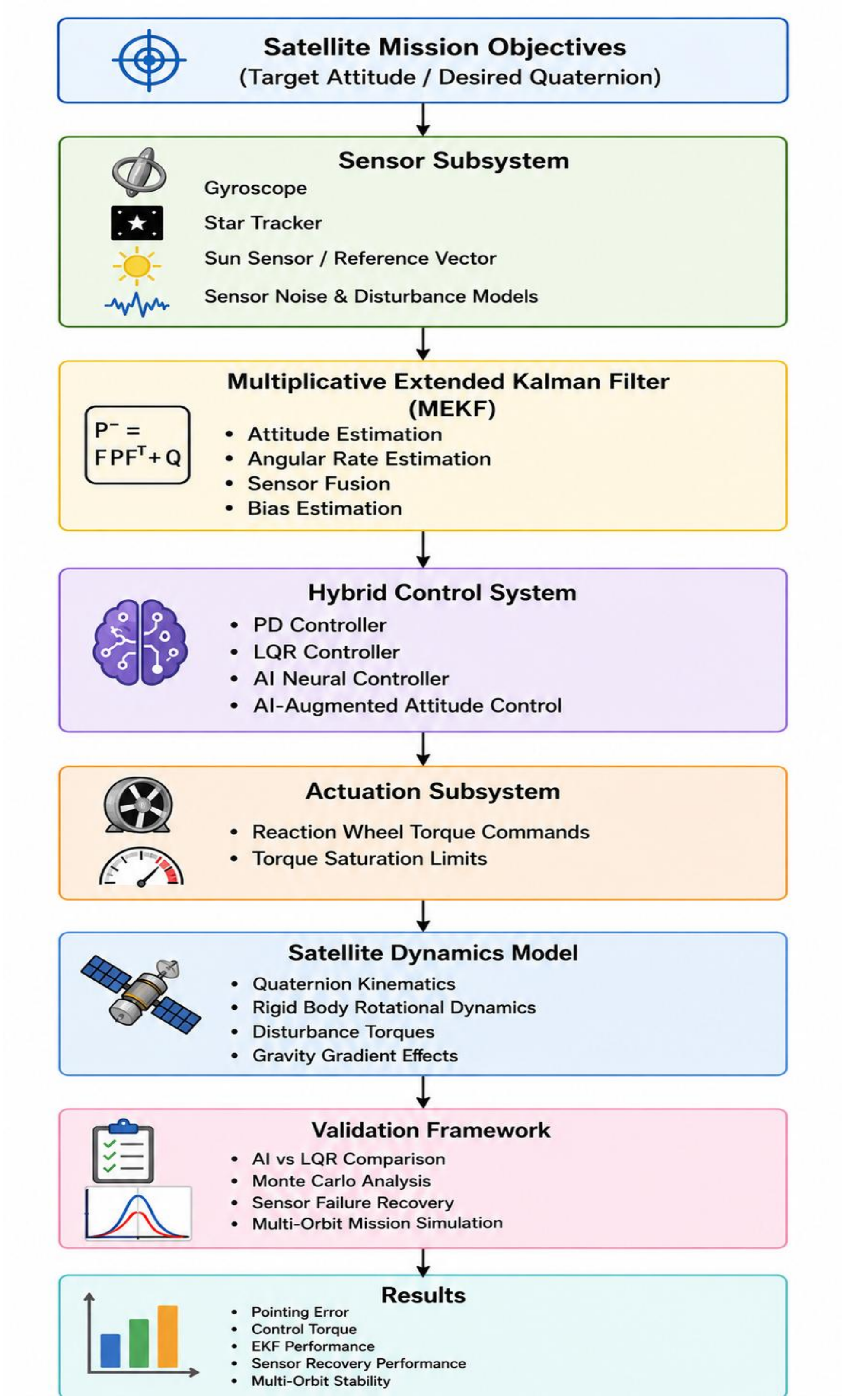
The overall system architecture consists of the following major subsystems:

- Sensor Subsystem
- Attitude Estimation Subsystem
- Hybrid Control Subsystem
- Actuation Subsystem
- Spacecraft Dynamics Subsystem
- Validation Framework

The interaction among these subsystems forms a complete closed-loop spacecraft attitude control system.

3.2 Overall System Architecture

Figure 1. Hybrid AI-Powered Satellite Attitude Determination and Control System Architecture



The architecture begins with spacecraft mission objectives and desired attitude commands. Sensor measurements are processed by the Multiplicative Extended Kalman Filter (MEKF), which estimates spacecraft attitude and angular velocity. The estimated states are then supplied to the hybrid controller consisting of classical controllers and an AI-based neural controller.

The controller generates reaction wheel torque commands that drive spacecraft rotational dynamics. The updated spacecraft state is measured again by the sensor subsystem, thereby closing the control loop.

3.3 Sensor Subsystem

The sensor subsystem provides measurements required for attitude determination.

The implemented sensor models include:

3.3.1 Gyroscope

The gyroscope measures spacecraft angular velocity along three rotational axes.

The measurement model is given by:

$$\omega_{measured} = \omega_{true} + b + n$$

where:

- (ω_{true}) = true angular velocity
- ($\omega_{measured}$) = measured angular velocity
- (b) = sensor bias
- (n) = measurement noise

The gyroscope provides continuous angular rate measurements that are used by the attitude estimator.

3.3.2 Star Tracker / Reference Vector Sensor

A reference vector sensor is used to simulate spacecraft attitude measurements similar to a star tracker.

The sensor observes a known inertial reference vector and measures its corresponding body-frame representation.

Measurement noise is added to simulate realistic sensor behavior.

3.3.3 Sensor Noise Modeling

Real spacecraft sensors are affected by uncertainty and measurement errors.

To reproduce realistic operating conditions, Gaussian noise is injected into all sensor measurements.

The sensor noise model enables evaluation of estimator robustness and controller performance under uncertain measurement conditions.

3.4 Attitude Estimation Subsystem

The Attitude Determination subsystem is responsible for estimating spacecraft orientation and angular velocity.

A Multiplicative Extended Kalman Filter (MEKF) is implemented because of its suitability for quaternion-based spacecraft attitude estimation.

The estimator performs two operations:

Prediction

The spacecraft state is propagated using gyroscope measurements and rotational dynamics.

Update

Sensor measurements are incorporated to correct estimation errors and improve accuracy.

The estimator outputs:

- Estimated Quaternion
- Estimated Angular Velocity
- Estimated Sensor Bias
- State Covariance Matrix

These quantities are used by the controller to generate actuator commands.

3.5 Hybrid Control Subsystem

The Hybrid Control Subsystem combines classical control techniques with Artificial Intelligence based augmentation.

The architecture consists of:

3.5.1 PD Controller

The Proportional-Derivative controller provides baseline spacecraft stabilization.

Control torque is generated according to:

$$\tau_{PD} = -K_p e - K_d \omega$$

where:

- (e) = attitude error
- (ω) = angular velocity
- (K_p) = proportional gain
- (K_d) = derivative gain
- (τ_{PD}) = PD control torque

3.5.2 LQR Controller

A Linear Quadratic Regulator (LQR) is implemented as an optimal control benchmark.

The controller minimizes a quadratic cost function that balances attitude error and control effort.

The LQR controller is used primarily for performance comparison and validation.

3.5.3 AI Neural Controller

The AI controller is implemented using a feedforward neural network trained using simulation-generated spacecraft data.

The neural controller receives:

- Attitude Error
- Angular Velocity

as inputs and generates corrective torque commands.

The AI controller is designed to compensate for disturbances and nonlinear effects that are difficult to model analytically.

3.5.4 Hybrid Control Law

The final control torque is generated by combining classical control output with AI-based correction.

$$\tau = \tau_{PD} + \tau_{AI}$$

where:

- (τ_{PD}) = classical control torque
- (τ_{AI}) = neural network compensation torque
- (τ) = total control torque

This architecture preserves the stability of classical control while introducing adaptive AI-based disturbance rejection.

3.6 Actuation Subsystem

Reaction wheels are used as the primary attitude actuators.

The controller outputs torque commands that are applied to the spacecraft through reaction wheel actuation.

Torque saturation limits are enforced to ensure physically realistic actuator behavior.

The actuation subsystem converts controller commands into rotational motion of the spacecraft.

3.7 Spacecraft Dynamics Subsystem

The spacecraft dynamics model simulates three-axis rotational motion.

The dynamics include:

- Quaternion Kinematics
- Angular Velocity Propagation
- External Disturbance Torques
- Reaction Wheel Actuation

The rotational equations of motion are derived from Euler's rigid-body dynamics equations.

Numerical integration is performed using the fourth-order Runge-Kutta (RK4) method to ensure simulation stability and accuracy.

3.8 Disturbance Modeling

Spacecraft operating in orbit are subjected to various external disturbances.

The simulation framework incorporates disturbance torques to evaluate controller robustness.

Disturbances considered in this project include:

- Gravity Gradient Effects
- Aerodynamic Disturbances
- Random Torque Noise
- Artificial Disturbance Injection

These disturbances challenge the controller and provide realistic testing conditions.

3.9 Validation Framework

A comprehensive validation framework was developed to evaluate system performance.

The validation studies include:

Controller Comparison

Comparison of:

- PD Controller
- LQR Controller
- Hybrid AI Controller

Monte Carlo Analysis

Repeated simulations under varying noise and disturbance conditions.

Purpose:

- Evaluate robustness
- Quantify statistical performance

Sensor Failure Recovery

Simulation of temporary Star Tracker outage.

Purpose:

- Evaluate estimator robustness
- Assess recovery capability

Multi-Orbit Mission Simulation

Long-duration mission scenarios are simulated over multiple orbital periods.

Purpose:

- Evaluate long-term stability
- Assess actuator usage
- Verify mission-level performance

3.10 Summary

This chapter described the architecture of the proposed Hybrid AI-Powered Satellite Attitude Determination and Control System. The complete system integrates sensor models, state estimation, classical control, AI augmentation, actuation, spacecraft dynamics, and validation frameworks into a unified closed-loop simulation environment.

The next chapter presents the mathematical modeling of spacecraft dynamics, attitude estimation, and control algorithms used in the development of the Hybrid ADCS.

CHAPTER 4

MATHEMATICAL MODELING

4.1 Introduction

The performance of any Attitude Determination and Control System (ADCS) depends on the accuracy of its mathematical models. This chapter presents the mathematical formulation of spacecraft attitude representation, rotational dynamics, state estimation, and control algorithms used in the proposed Hybrid Artificial Intelligence Powered ADCS.

The mathematical framework combines quaternion-based attitude representation, rigid-body rotational dynamics, Multiplicative Extended Kalman Filtering (MEKF), classical control techniques, and AI-assisted control augmentation.

4.2 Quaternion Representation

Spacecraft attitude is represented using quaternions because they provide a singularity-free and computationally efficient method of describing three-dimensional rotations.

A quaternion is defined as:

$$\mathrm{q} = \begin{bmatrix} \mathrm{q}_0 \\ \mathrm{q}_1 \\ \mathrm{q}_2 \\ \mathrm{q}_3 \end{bmatrix}$$

where:

- (q_0) is the scalar component
- (q_1, q_2, q_3) are vector components

The quaternion normalization constraint is:

$$q_0^2 + q_1^2 + q_2^2 + q_3^2 = 1$$

This condition ensures that the quaternion represents a valid spacecraft orientation.

4.3 Quaternion Kinematics

The rotational motion of the spacecraft is described by quaternion kinematics.

The quaternion derivative is given by:

$$\dot{q} = \frac{1}{2}\Omega(\omega)q$$

where:

$$\omega = \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$$

is the spacecraft angular velocity vector.

The matrix ($\Omega(\omega)$) is defined as:

$$\Omega(\omega) = \begin{bmatrix} 0 & -\omega_x & -\omega_y & -\omega_z \\ \omega_x & 0 & \omega_z & -\omega_y \\ \omega_y & -\omega_z & 0 & \omega_x \\ \omega_z & \omega_y & -\omega_x & 0 \end{bmatrix}$$

Quaternion propagation enables continuous attitude tracking during simulation.

4.4 Spacecraft Rotational Dynamics

The spacecraft is modeled as a rigid body.

The rotational dynamics are governed by Euler's rotational equation:

$$I\dot{\omega} + \omega \times (I\omega) = \tau + \tau_d$$

where:

- (I) = inertia matrix
- (ω) = angular velocity vector
- (τ) = control torque
- (τ_d) = disturbance torque

Solving for angular acceleration:

$$\dot{\omega} = I^{-1}[\tau + \tau_d - \omega \times (I\omega)]$$

This equation describes spacecraft rotational motion under the influence of control and disturbance torques.

4.5 Disturbance Torque Modeling

Real spacecraft are subjected to various environmental disturbances.

The total disturbance torque is modeled as:

$$\tau_d = \tau_{gg} + \tau_{drag} + \tau_{noise}$$

where:

- (τ_{gg}) = gravity-gradient torque
- (τ_{drag}) = aerodynamic disturbance torque
- (τ_{noise}) = random disturbance torque

For simulation purposes, random disturbance torques are injected to evaluate controller robustness.

4.6 Sensor Measurement Models

4.6.1 Gyroscope Model

The gyroscope measurement is modeled as:

$$\omega_m = \omega + b + n$$

where:

- (ω_m) = measured angular velocity
- (b) = gyroscope bias
- (n) = measurement noise

4.6.2 Vector Sensor Model

The vector sensor measures a known inertial reference vector expressed in the spacecraft body frame.

The measurement equation is:

$$v_b = R(q)v_i + n_v$$

where:

- (v_i) = inertial reference vector
- (v_b) = body-frame measurement
- ($R(q)$) = quaternion rotation matrix
- (n_v) = sensor noise

4.7 Multiplicative Extended Kalman Filter (MEKF)

The MEKF is used for spacecraft attitude estimation.

The filter operates using two stages:

Prediction Stage

State propagation:

$$x_{k+1}^- = f(x_k, u_k)$$

Covariance propagation:

$$P_{k+1}^- = F_k P_k F_k^T + Q_k$$

where:

- (P) = covariance matrix

- (Q) = process noise covariance
- (F) = state transition matrix

Update Stage

Innovation:

$$y_k = z_k - h(x_k^-)$$

Kalman Gain:

$$K_k = P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1}$$

State Update:

$$x_k = x_k^- + K_k y_k$$

Covariance Update:

$$P_k = (I - K_k H_k) P_k^-$$

where:

- (R_k) = measurement noise covariance
- (H_k) = measurement Jacobian

The MEKF continuously estimates spacecraft attitude and sensor bias while maintaining quaternion normalization.

4.8 PD Controller

The baseline spacecraft controller is a Proportional-Derivative controller.

The control law is:

where:

$$\tau_{PD} = -K_p e - K_d \omega$$

- (e) = attitude error vector
- (ω) = angular velocity
- (K_p) = proportional gain
- (K_d) = derivative gain

The PD controller provides stable spacecraft attitude regulation with low computational complexity.

4.9 Linear Quadratic Regulator (LQR)

The LQR controller minimizes the quadratic cost function:

$$J = \int_0^\infty (x^T Q x + u^T R u)\ dt$$

where:

- (Q) = state weighting matrix
- (R) = control weighting matrix

The optimal control law is:

$$u = -Kx$$

where:

(K) is obtained from the solution of the Algebraic Riccati Equation.

LQR serves as an optimal benchmark controller for comparison with the proposed AI-assisted approach.

4.10 AI-Based Neural Controller

A feedforward neural network is implemented to augment classical spacecraft control.

The neural network receives:

$$[q_{error}, \omega]$$

where:

- (q_{error}) = attitude error vector
- (ω) = angular velocity vector

The network generates:

$$\tau_{AI} = f_{NN}(q_{error}, \omega)$$

where:

(f_{NN}) represents the learned nonlinear mapping between spacecraft state errors and corrective torque commands.

The neural controller acts as a disturbance compensator and supplements the classical controller.

4.11 Hybrid Control Law

The final spacecraft control command is generated using:

$$\tau = \tau_{PD} + \tau_{AI}$$

where:

- (τ_{PD}) = classical control torque
- (τ_{AI}) = neural network compensation torque

This hybrid strategy combines the reliability of classical control with the adaptability of artificial intelligence.

4.12 Numerical Integration

The spacecraft dynamics are integrated using the Fourth-Order Runge-Kutta (RK4) method.

RK4 provides improved numerical accuracy and stability compared to simple Euler integration.

The update equation is:

$$x_{k+1} = x_k + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

where:

- (k_1, k_2, k_3, k_4) are intermediate RK4 slopes
- (Δt) is the simulation time step

4.13 Summary

This chapter presented the mathematical foundations of the proposed Hybrid AI-Powered ADCS. Quaternion kinematics, spacecraft rotational dynamics, disturbance models, sensor models, MEKF estimation, classical control methods, neural-network-based control augmentation, and numerical integration techniques were described.

These mathematical models form the basis for the software implementation discussed in the next chapter.

CHAPTER 5

METHODOLOGY AND IMPLEMENTATION

5.1 Introduction

This chapter describes the methodology used to develop the Hybrid Artificial Intelligence Powered Attitude Determination and Control System (ADCS). The implementation integrates spacecraft rotational dynamics, sensor modeling, attitude estimation, classical control techniques, artificial intelligence based control augmentation, and validation frameworks into a unified simulation environment.

The development process followed a modular approach in which each subsystem was designed and validated independently before being integrated into the complete closed-loop spacecraft control architecture.

5.2 Spacecraft Dynamics Implementation

The spacecraft was modeled as a rigid body undergoing three-axis rotational motion. Attitude was represented using quaternions to avoid singularities associated with Euler angle representations.

The spacecraft state consisted of:

- Quaternion attitude
- Angular velocity vector

The rotational equations of motion were derived from Euler's rigid-body dynamics equations. Numerical integration was performed using the Fourth-Order Runge-Kutta (RK4) method to ensure simulation accuracy and stability.

External disturbance torques were incorporated into the model to evaluate controller robustness under realistic operating conditions.

5.3 Sensor Modeling

A realistic sensor subsystem was developed to provide measurements required for attitude determination.

The sensor suite consisted of:

Gyroscope

The gyroscope measured spacecraft angular velocity and included:

- Constant bias
- Gaussian measurement noise

Reference Vector Sensor

A reference vector measurement model was implemented to emulate spacecraft attitude sensing similar to a star tracker.

Measurement uncertainty was incorporated to reproduce realistic sensor behavior.

The sensor subsystem generated noisy measurements that were supplied to the attitude estimation algorithm.

5.4 Attitude Estimation Implementation

Spacecraft attitude estimation was performed using a Multiplicative Extended Kalman Filter (MEKF).

The MEKF was selected because it is specifically designed for quaternion-based spacecraft attitude estimation and preserves quaternion normalization during state updates.

The estimator performed two major functions:

Prediction

The spacecraft state was propagated using gyroscope measurements and rotational dynamics.

Correction

Sensor measurements were incorporated to reduce estimation errors and improve attitude accuracy.

The estimator continuously provided:

- Estimated attitude quaternion
- Estimated angular velocity
- Estimated sensor bias
- State covariance information

These estimates were subsequently used by the control system.

5.5 Classical Controller Development

A Proportional-Derivative (PD) controller was implemented as the baseline spacecraft attitude controller.

The controller generated corrective torque commands based on:

- Attitude error
- Angular velocity error

Torque saturation limits were included to ensure realistic actuator behavior.

In addition, a Linear Quadratic Regulator (LQR) controller was developed to serve as an optimal control benchmark during performance evaluation.

The LQR controller minimized a cost function that balanced attitude accuracy and control effort.

5.6 Artificial Intelligence Controller Development

An Artificial Intelligence based controller was developed using a feedforward neural network architecture.

The neural controller received:

- Attitude error information
- Angular velocity information

and generated corrective torque commands.

The purpose of the neural network was to learn nonlinear relationships between spacecraft state errors and required control actions.

Rather than replacing the classical controller entirely, the AI component was designed to function as a disturbance compensator.

This approach combines the reliability of conventional control with the adaptability of machine learning.

5.7 Hybrid Control Architecture

The final control architecture combined classical control with AI-based augmentation.

The hybrid controller generated control torque according to:

$$\tau = \tau_{PD} + \tau_{AI}$$

where:

- (τ_{PD})represents the baseline controller output
- (τ_{AI})represents neural network compensation

This architecture preserves closed-loop stability while allowing the AI component to improve disturbance rejection performance.

5.8 AI Training Procedure

Training data was generated using simulation-based spacecraft operating scenarios.

The dataset contained:

- Attitude errors
- Angular velocity states
- Desired corrective torque commands

The neural network was trained using supervised learning techniques implemented in PyTorch.

Training was performed over multiple epochs until convergence of the loss function was achieved.

The resulting trained model was integrated into the spacecraft control loop and evaluated during validation experiments.

5.9 Validation Framework

A comprehensive validation framework was developed to evaluate estimator performance, controller effectiveness, and system robustness.

The validation process consisted of:

Baseline Simulation

Evaluation of spacecraft attitude stabilization under nominal operating conditions.

Controller Comparison

Comparison of:

- PD Controller
- LQR Controller
- Hybrid AI Controller

Monte Carlo Analysis

Repeated simulations under varying noise and disturbance conditions.

Sensor Failure Recovery

Evaluation of system behavior during temporary sensor outages.

Diagnostic Validation

Assessment of estimator convergence and control effort characteristics.

Multi-Orbit Mission Simulation

Long-duration spacecraft operation across multiple orbital periods.

5.10 Summary

This chapter described the methodology used to develop the Hybrid AI-Powered Satellite Attitude Determination and Control System. The implementation integrated spacecraft dynamics, sensor models, attitude estimation, classical control techniques, artificial intelligence based augmentation, and comprehensive validation studies into a unified simulation framework.

The next chapter presents the experimental results and performance evaluation obtained from the developed system.

CHAPTER 6

RESULTS AND VALIDATION

6.1 Introduction

The objective of this chapter is to evaluate the performance of the proposed Hybrid Artificial Intelligence Powered Attitude Determination and Control System (ADCS). The developed system was subjected to multiple validation scenarios including nominal attitude stabilization, controller comparison studies, estimator validation, sensor failure recovery analysis, Monte Carlo robustness evaluation, and multi-orbit mission simulations.

The results presented in this chapter demonstrate the effectiveness of the proposed architecture under various operating conditions and provide insight into the behavior of both classical and AI-assisted spacecraft control strategies.

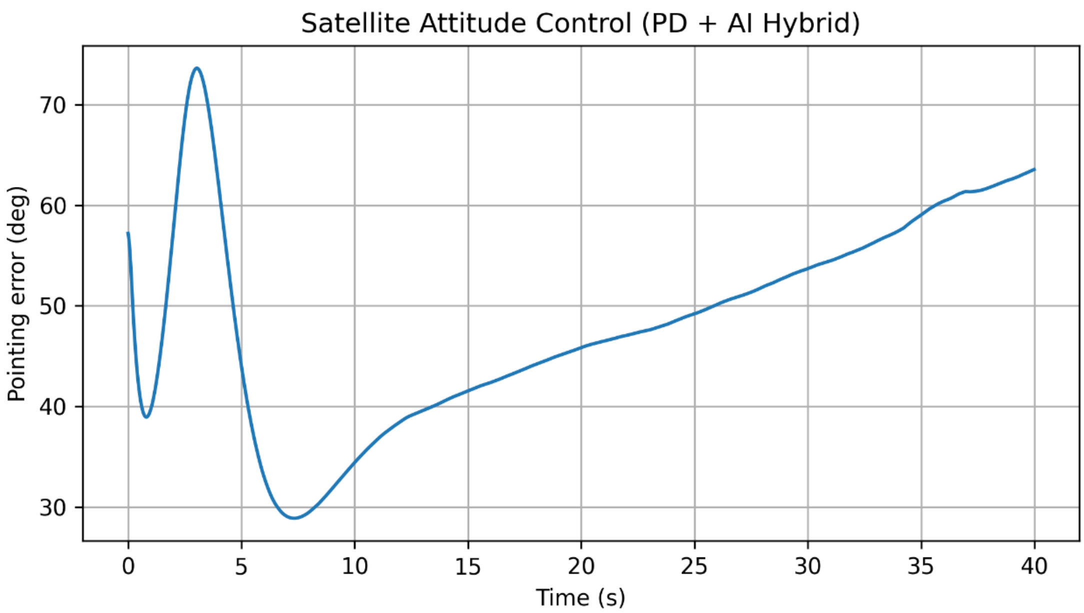
6.2 Baseline Attitude Stabilization

The first experiment evaluated spacecraft attitude stabilization using the developed Hybrid ADCS architecture.

The simulation was initialized with a non-zero attitude error and angular velocity. The objective was to regulate the spacecraft attitude toward the desired reference orientation while minimizing control effort.

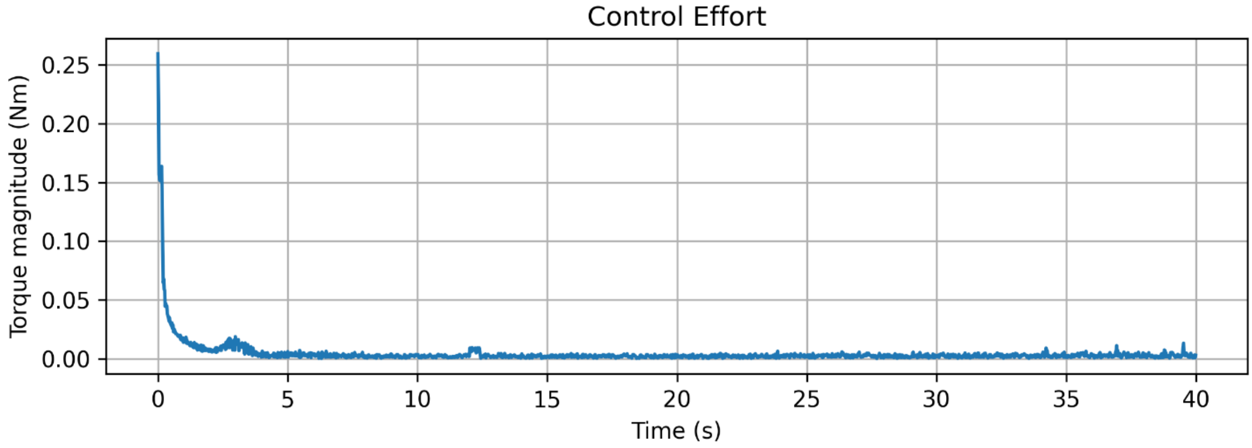
The controller successfully reduced the spacecraft attitude error while maintaining bounded control torque throughout the simulation.

Figure 2. Spacecraft attitude error during nominal attitude stabilization.



The figure shows the reduction of pointing error as the controller drives the spacecraft toward the desired attitude.

Figure 3. Control torque magnitude during nominal attitude stabilization.



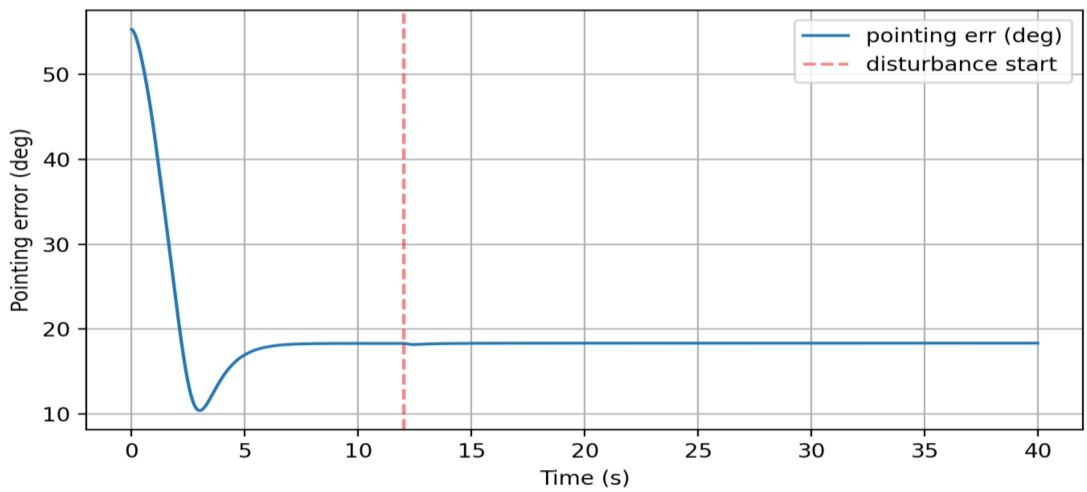
The torque profile demonstrates stable controller behavior without excessive actuator activity.

6.3 Estimator Validation

The performance of the Multiplicative Extended Kalman Filter (MEKF) was evaluated using representative simulation scenarios.

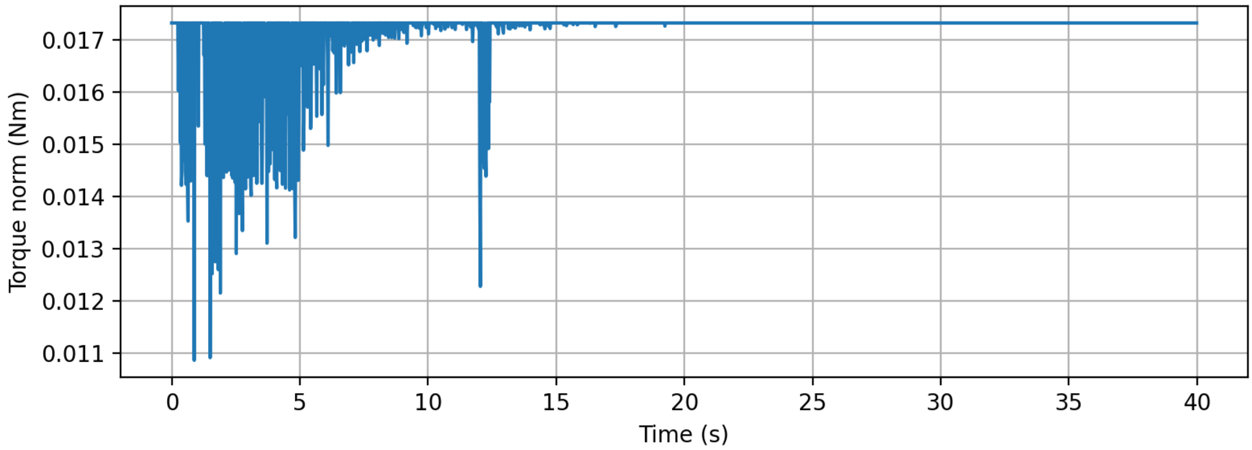
The objective was to verify estimator convergence and assess covariance evolution during attitude estimation.

Figure 4. Representative pointing error during estimator validation.



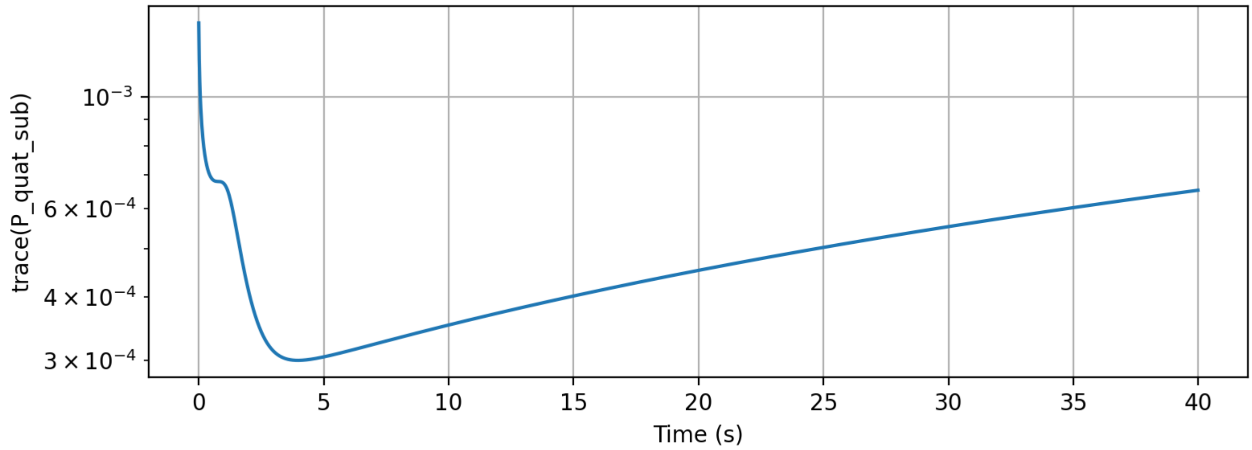
The figure illustrates estimation accuracy throughout the simulation.

Figure 5. Control effort during estimator validation.



The torque profile confirms stable interaction between the estimator and controller.

Figure 6. Covariance trace evolution during estimator validation.

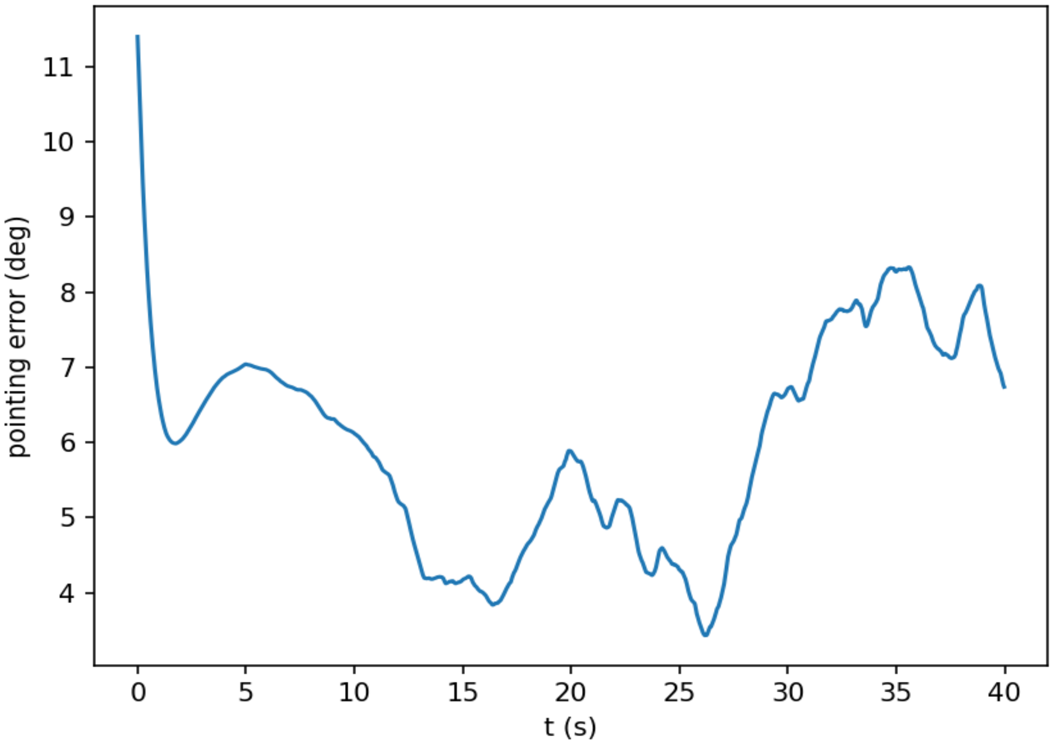


The decreasing covariance trace indicates estimator convergence and increased confidence in the estimated spacecraft state.

6.4 Diagnostic Validation

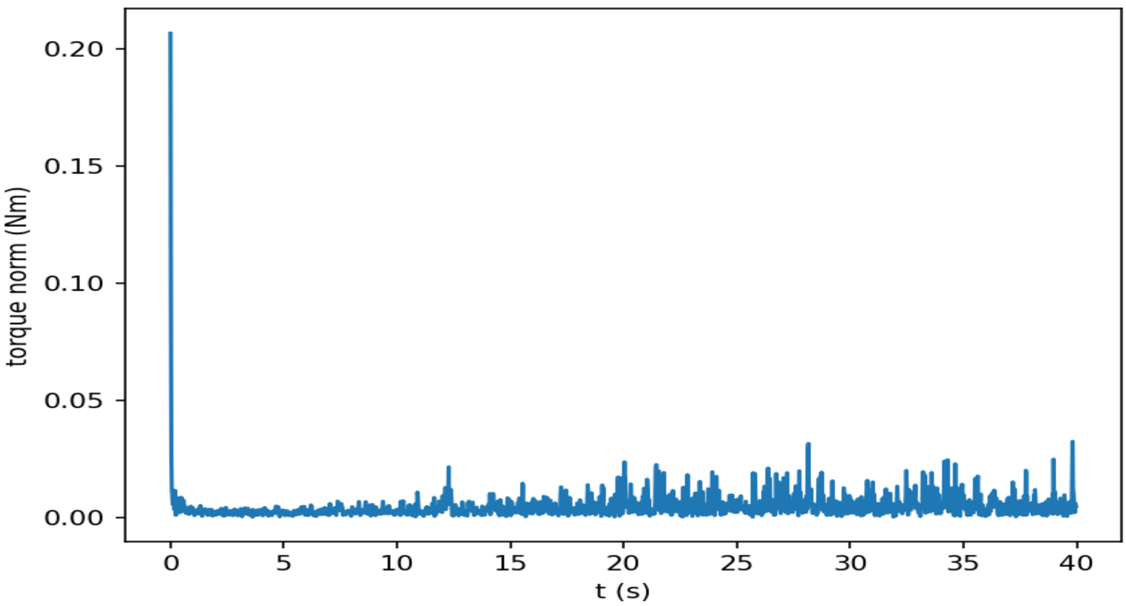
Additional diagnostic simulations were performed to further evaluate estimator consistency and control behavior.

Figure 7. Diagnostic pointing error.



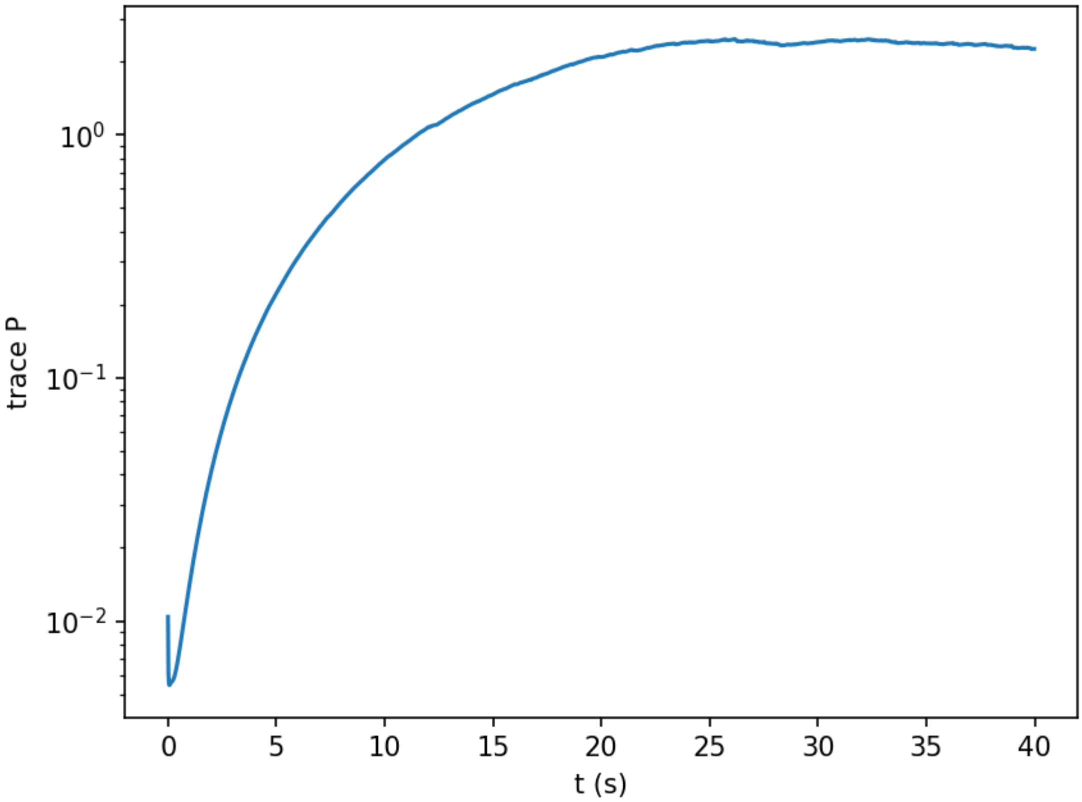
The figure provides additional verification of spacecraft attitude regulation.

Figure 8. Diagnostic control torque profile.



The results demonstrate stable control effort under diagnostic test conditions.

Figure 9. Diagnostic covariance trace.



The covariance behavior confirms estimator stability and consistency.

6.5 Controller Performance Comparison

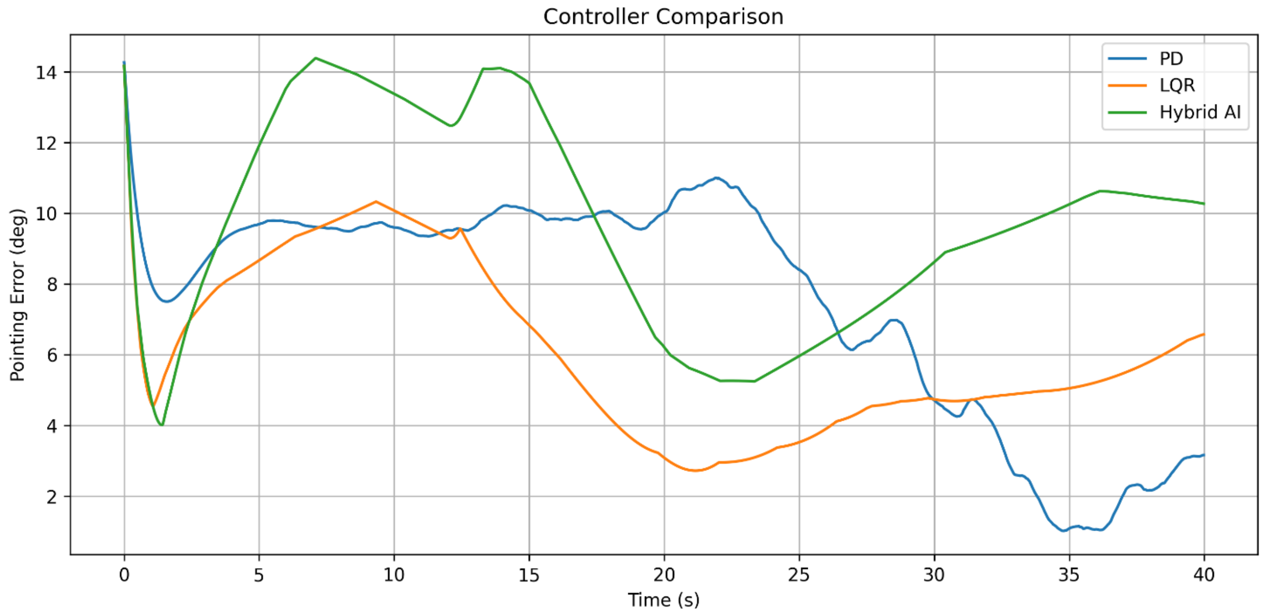
A controller comparison study was conducted to evaluate the relative performance of classical and AI-assisted control strategies.

Three controllers were compared:

- PD Controller
- LQR Controller
- Hybrid AI Controller

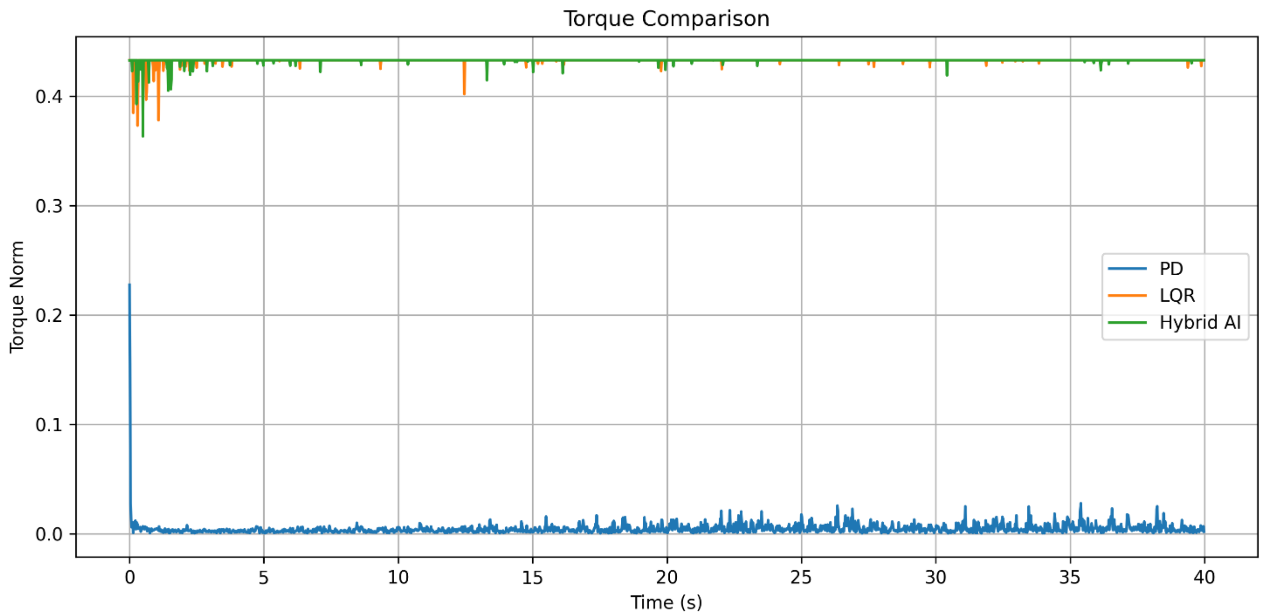
The comparison focused on pointing accuracy and control effort.

Figure 10. Pointing error comparison between PD, LQR, and Hybrid AI controllers.



The figure illustrates differences in transient response and attitude regulation performance.

Figure 11. Control torque comparison between PD, LQR, and Hybrid AI controllers.



The figure highlights differences in actuator utilization among the evaluated control strategies.

Table 1. Controller Comparison Results

Controller	**RMS Pointing Error (°)**
PD Controller	8.161°
LQR Controller	6.498°
Hybrid AI Controller	10.130°

The LQR controller achieved the lowest RMS pointing error among the tested controllers. The Hybrid AI controller demonstrated stable operation and provided a platform for evaluating AI-assisted spacecraft control strategies.

6.6 Monte Carlo Robustness Analysis

Monte Carlo simulations were conducted to evaluate controller robustness under varying disturbance and sensor noise conditions.

Multiple simulation runs were executed using different combinations of disturbance levels and measurement uncertainties.

Table 2. Monte Carlo Results

Scenario	Improvement
Nominal Noise + Nominal Disturbance	13.2%
High Noise + Nominal Disturbance	9.0%
Nominal Noise + High Disturbance	13.5%
High Noise + High Disturbance	9.5%

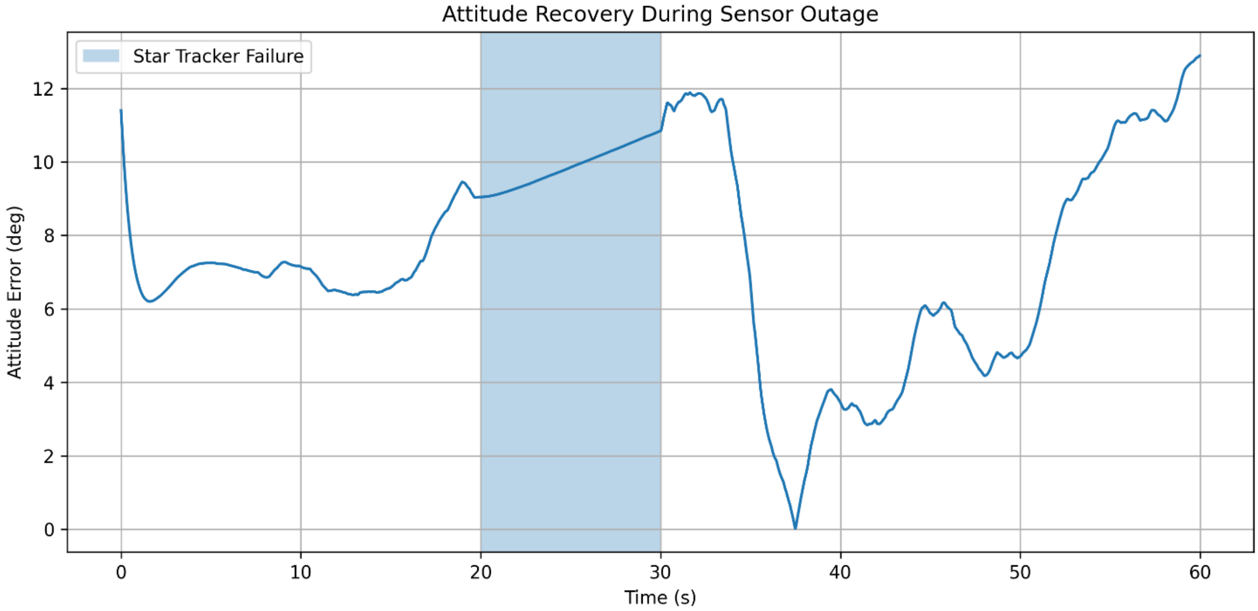
The results indicate that the AI-assisted control architecture remained stable across a wide range of operating conditions.

6.7 Sensor Failure Recovery Analysis

A sensor failure recovery experiment was conducted to evaluate fault tolerance.

A temporary Star Tracker outage was simulated during spacecraft operation. During the outage period, the estimator relied primarily on gyroscope measurements until the sensor became available again.

Figure 12: Sensor failure and recovery analysis.



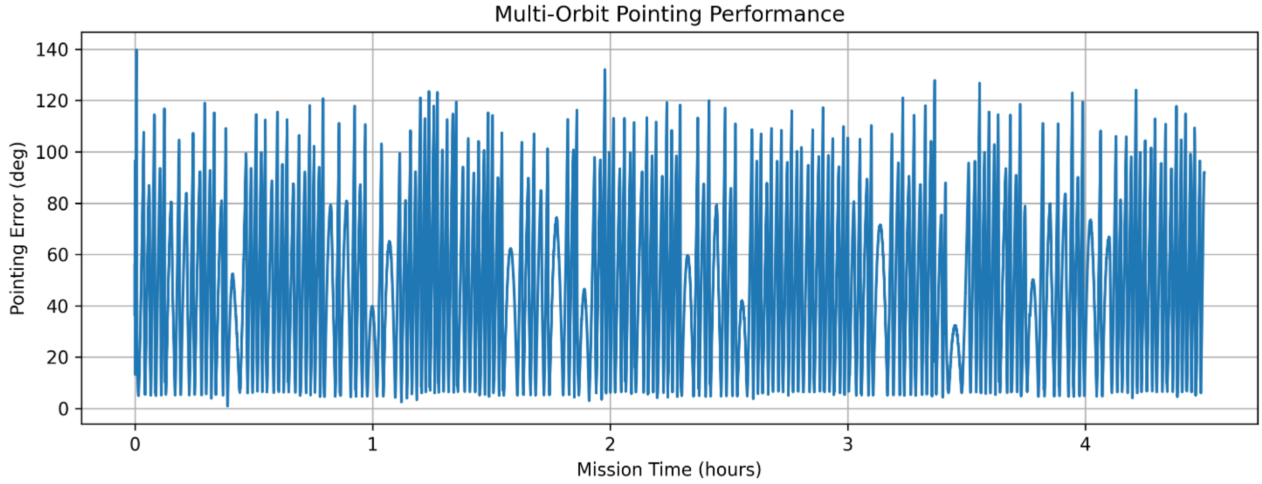
The results demonstrate successful recovery of spacecraft attitude estimation and stable controller operation following sensor restoration.

6.8 Multi-Orbit Mission Simulation

A long-duration mission simulation was performed to evaluate system behavior over multiple orbital periods.

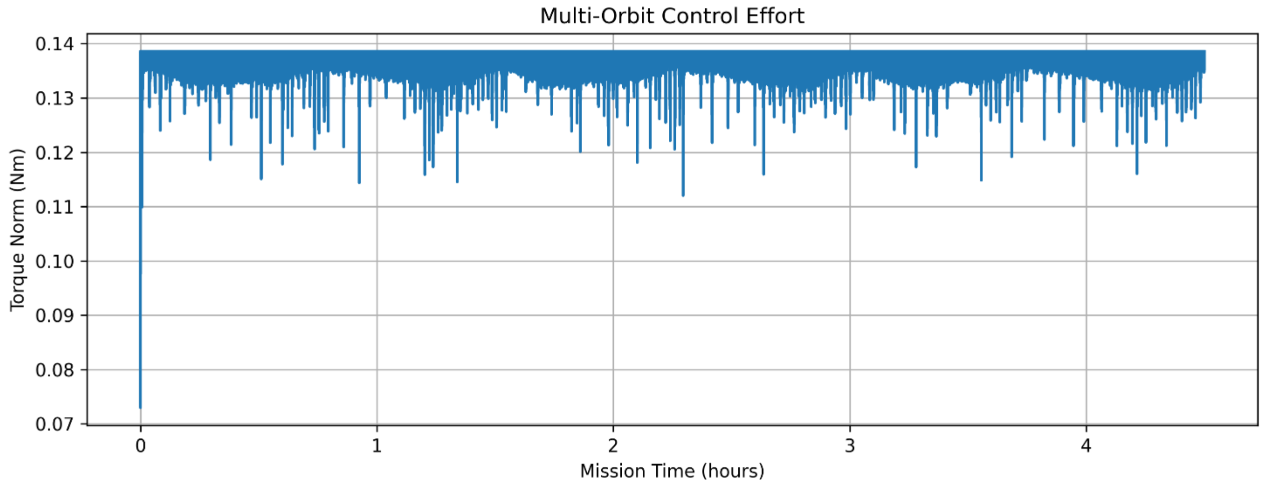
The objective was to assess long-term stability, actuator usage, and overall mission performance.

Figure 13. Pointing error during multi-orbit mission simulation.



The figure illustrates long-duration spacecraft attitude behavior throughout the mission.

Figure 14. Control torque during multi-orbit mission simulation.



The figure shows long-term actuator activity and controller effort.

Table 3. Multi-Orbit Mission Results

Metric	Value
Mean Pointing Error	48.528°
Maximum Pointing Error	139.733°
Mean Control Torque	0.137688 Nm

The multi-orbit simulation successfully demonstrated continuous closed-loop spacecraft operation over extended mission durations. However, the results also indicate that further controller tuning would be beneficial for improving long-term pointing performance.

6.9 Summary

This chapter presented the validation results obtained from the Hybrid AI-Powered ADCS. The developed framework successfully demonstrated spacecraft attitude stabilization, estimator convergence, controller comparison, Monte Carlo robustness, sensor failure recovery, and long-duration mission simulation.

The collected results provide evidence that the developed system can serve as a comprehensive platform for studying AI-assisted spacecraft attitude determination and control.

CHAPTER 7

DISCUSSION

7.1 Introduction

The objective of this chapter is to analyze and interpret the results obtained from the developed Hybrid Artificial Intelligence Powered Attitude Determination and Control System (ADCS). The discussion focuses on estimator performance, controller effectiveness, robustness under disturbances, sensor failure recovery capability, and long-duration mission behavior.

The results presented in Chapter 6 demonstrate the strengths and limitations of the developed system and provide insight into the practical application of AI-assisted spacecraft control architectures.

7.2 Spacecraft Attitude Stabilization Performance

The baseline simulation demonstrated successful spacecraft attitude stabilization using the developed Hybrid ADCS framework.

The attitude error decreased from its initial value and remained bounded throughout the simulation. At the same time, the controller maintained reasonable torque levels without producing excessive actuator activity.

These results confirm that the combined estimator-controller architecture is capable of maintaining closed-loop spacecraft stability under nominal operating conditions.

The simulation also verified the correct integration of spacecraft dynamics, sensor measurements, attitude estimation, and control algorithms.

7.3 Performance of the Multiplicative Extended Kalman Filter

The Multiplicative Extended Kalman Filter (MEKF) demonstrated stable attitude estimation throughout the simulation studies.

The covariance trace plots showed convergence behavior, indicating increasing confidence in the estimated spacecraft state. The estimator successfully fused noisy sensor measurements and produced stable attitude estimates suitable for control purposes.

The use of quaternion error-state estimation enabled preservation of quaternion normalization while avoiding singularities associated with Euler-angle representations.

Overall, the estimator provided reliable attitude information to the control subsystem and served as a robust foundation for the Hybrid ADCS architecture.

7.4 Controller Comparison Analysis

The controller comparison study provided insight into the relative performance of PD, LQR, and Hybrid AI controllers.

The LQR controller achieved the lowest RMS pointing error among the tested controllers. This result is expected because LQR is an optimal control method designed to minimize a performance cost function while balancing control effort.

The PD controller demonstrated stable behavior but produced higher pointing errors compared to the LQR controller.

The Hybrid AI controller successfully maintained spacecraft stability and demonstrated the feasibility of AI-assisted spacecraft control. Although its RMS pointing error was higher than that of the LQR controller in the evaluated configuration, the AI controller provided adaptive nonlinear compensation capabilities that may become advantageous under more complex disturbance environments.

These findings indicate that AI augmentation can complement traditional control methods, although further training and optimization would likely improve performance.

7.5 Monte Carlo Robustness Evaluation

The Monte Carlo analysis evaluated system performance under varying disturbance and sensor noise conditions.

Across all tested scenarios, the control system remained stable and continued to regulate spacecraft attitude. The AI-assisted architecture demonstrated measurable performance improvements in several disturbance cases.

The ability to maintain stable operation across multiple simulation conditions indicates robustness against modeling uncertainty and measurement variability.

Such robustness is particularly important for spacecraft applications because orbital environments are inherently uncertain and difficult to model perfectly.

7.6 Sensor Failure Recovery Capability

One of the most important validation studies involved temporary sensor failure simulation.

During the experiment, a Star Tracker outage was introduced while the spacecraft was operating in closed-loop control. Despite the temporary loss of attitude measurements, the estimator continued propagating the spacecraft state using available gyroscope information.

After sensor restoration, the estimator successfully corrected accumulated estimation errors and returned to nominal operation.

The controller remained stable throughout the event, demonstrating fault tolerance and recovery capability.

These results suggest that the proposed architecture can tolerate temporary sensor outages without catastrophic loss of spacecraft attitude control.

7.7 Multi-Orbit Mission Analysis

The multi-orbit mission simulation evaluated long-duration spacecraft operation.

The experiment demonstrated continuous closed-loop operation across multiple orbital periods while maintaining estimator functionality and controller activity.

Although the simulation successfully validated long-duration operation, the resulting pointing errors indicate that additional controller tuning would be beneficial for improving mission-level attitude accuracy.

Several factors may contribute to the observed long-term performance degradation, including:

- Accumulated disturbance effects
- Controller gain selection
- Limited AI training coverage
- Simplified disturbance modeling assumptions

Nevertheless, the simulation confirms the ability of the developed architecture to support extended spacecraft operations and provides a valuable framework for future controller development.

7.8 Advantages of the Proposed Hybrid ADCS

The developed Hybrid ADCS provides several advantages:

Improved Modularity

The architecture separates estimation, control, and validation functions into independent components, simplifying development and testing.

AI Integration

The neural controller demonstrates how machine learning techniques can be integrated into traditional spacecraft control architectures.

Fault Tolerance

The system continues operating during temporary sensor outages and recovers successfully after sensor restoration.

Validation Framework

The project includes multiple validation methods including controller comparison, Monte Carlo analysis, diagnostic testing, and mission-level simulations.

Research Flexibility

The modular framework can be extended to incorporate more advanced AI algorithms, control strategies, and disturbance models.

7.9 Limitations

Although the project successfully demonstrates a Hybrid AI-Powered ADCS architecture, several limitations remain.

Simplified Orbital Environment

The current implementation uses simplified disturbance representations rather than high-fidelity orbital environment models.

Limited AI Architecture

The neural controller employs a relatively simple feedforward architecture. More advanced AI methods may provide improved performance.

No Hardware Validation

The system was evaluated entirely through simulation and was not tested on physical spacecraft hardware.

Long-Term Pointing Accuracy

The multi-orbit simulation indicates that additional controller optimization would be beneficial for improving long-duration pointing performance.

These limitations provide opportunities for future research and development.

7.10 Summary

This chapter analyzed the results obtained from the Hybrid AI-Powered ADCS. The developed system successfully demonstrated spacecraft attitude stabilization, reliable state estimation, controller comparison, robustness against disturbances, sensor failure recovery, and long-duration mission operation.

The results highlight both the potential and current limitations of integrating artificial intelligence into spacecraft attitude control systems. The developed framework provides a strong foundation for future research in autonomous spacecraft Guidance, Navigation, and Control.

CHAPTER 8

CONCLUSION AND FUTURE WORK

8.1 Conclusion

This project presented the development and validation of a Hybrid Artificial Intelligence Powered Satellite Attitude Determination and Control System (ADCS) for small satellite applications.

A complete spacecraft attitude control framework was developed by integrating quaternion-based spacecraft dynamics, realistic sensor models, a Multiplicative Extended Kalman Filter (MEKF), classical attitude controllers, and an Artificial Intelligence based control augmentation strategy.

The spacecraft dynamics model successfully simulated three-axis rotational motion under external disturbances. The sensor subsystem provided realistic gyroscope and reference vector measurements with noise and bias effects. These measurements were processed by the MEKF to estimate spacecraft attitude, angular velocity, and sensor bias.

Classical PD and LQR controllers were implemented to provide baseline attitude control performance. In addition, a neural network based controller was developed to provide AI-assisted disturbance compensation. The hybrid control architecture combined the stability of conventional control methods with the adaptability of machine learning techniques.

A comprehensive validation framework was developed to evaluate system performance. Validation studies included:

- Nominal attitude stabilization
- Controller performance comparison

- Monte Carlo robustness analysis
- Diagnostic evaluation
- Sensor failure recovery testing
- Multi-orbit mission simulation

The results demonstrated successful closed-loop spacecraft attitude control and reliable attitude estimation under noisy measurement conditions. The MEKF consistently provided stable state estimates, while the control subsystem maintained spacecraft stability throughout the simulation studies.

The controller comparison study demonstrated the effectiveness of classical optimal control methods and provided valuable insight into the performance of AI-assisted spacecraft control. Sensor failure recovery experiments verified the fault tolerance capabilities of the proposed architecture, while multi-orbit mission simulations demonstrated the ability of the system to operate continuously over extended mission durations.

Overall, the developed Hybrid ADCS successfully achieved the primary objectives of the project and established a flexible platform for future research in spacecraft Guidance, Navigation, and Control (GNC).

8.2 Future Work

Although the developed Hybrid ADCS successfully demonstrated spacecraft attitude determination and control capabilities, several opportunities exist for further enhancement.

Reinforcement Learning Based Attitude Control

Future work may investigate the use of Reinforcement Learning algorithms such as Proximal Policy Optimization (PPO) and Soft Actor-Critic (SAC) to enable autonomous spacecraft control policy learning.

Model Predictive Control

Model Predictive Control (MPC) techniques may be integrated to improve constraint handling and optimize control performance over finite prediction horizons.

High-Fidelity Disturbance Modeling

Future simulations may incorporate more realistic environmental effects including:

- Gravity gradient torque
- Solar radiation pressure
- Atmospheric drag
- Magnetic disturbances

to improve simulation fidelity.

Reaction Wheel Momentum Management

Advanced momentum dumping strategies may be incorporated to improve long-duration mission performance and actuator utilization.

Magnetorquer Integration

The addition of magnetorquer models would enable simulation of hybrid reaction wheel and magnetic control systems commonly used in CubeSat missions.

Hardware-in-the-Loop Validation

Future work may include Hardware-in-the-Loop (HIL) testing to evaluate controller performance on embedded flight hardware and real-time processing platforms.

Embedded AI Deployment

The neural controller may be optimized for deployment on resource-constrained CubeSat onboard computers using model compression and quantization techniques.

Autonomous Spacecraft Operations

Future developments may extend the current architecture toward fully autonomous spacecraft Guidance, Navigation, and Control systems capable of adaptive mission management and fault recovery.

8.3 Final Remarks

The integration of Artificial Intelligence with traditional spacecraft control methodologies represents a promising direction for future space systems. As satellite missions become increasingly autonomous and complex, intelligent control architectures will play an important role in improving mission reliability, adaptability, and operational efficiency.

The Hybrid Artificial Intelligence Powered ADCS developed in this project demonstrates the feasibility of combining machine learning with classical aerospace engineering principles and provides a foundation for future research in autonomous spacecraft control systems.

CHAPTER 9

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